Simulation task

1. Simulation of digital beamforming under user direction, beam width, SINR, antenna spacing, and channel condition (BL4)

Code

```
clc;
clear;
close all;

%% Parameters
N = 8;              % Number of antenna elements
fc = 2.4e9;         % Carrier frequency (Hz)
c = 3e8;            % Speed of light
lambda = c/fc;      % Wavelength
d = lambda/2;       % Antenna spacing

theta = -90:0.1:90;    % Observation angles
user_angle = 30;       % Desired user direction (degrees)

%% Steering Vector
n = 0:N-1;
w = exp(-1j*2*pi*d*n'*sind(user_angle)/lambda);

%% Array Response
AF = zeros(size(theta));

for k = 1:length(theta)
   a = exp(-1j*2*pi*d*n'*sind(theta(k))/lambda);
```

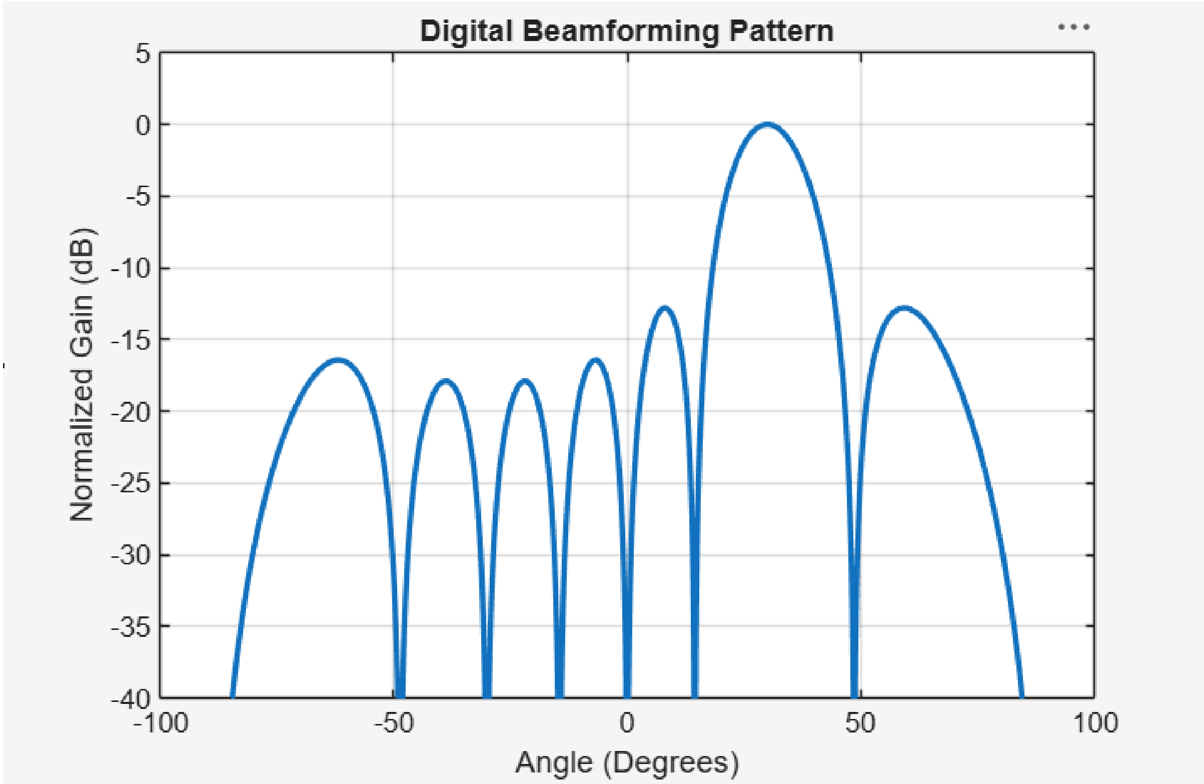
```
    AF(k) = abs(w' * a);
end

AF = AF/max(AF);

%% Beam Pattern
figure;
plot(theta,20*log10(AF),'LineWidth',2);
grid on;
xlabel('Angle (Degrees)');
ylabel('Normalized Gain (dB)');
title('Digital Beamforming Pattern');
ylim([-40 5]);

%% Beam Width
[maxGain,index] = max(AF);
halfPower = maxGain/sqrt(2);

left = find(AF(1:index)<=halfPower,1,'last');
right = index-1+find(AF(index:end)<=halfPower,1,'first');

beamwidth = theta(right)-theta(left);

fprintf('Beam Width = %.2f Degrees\n',beamwidth);

%% SINR Calculation
signalPower = 10;
interferencePower = 2;
```

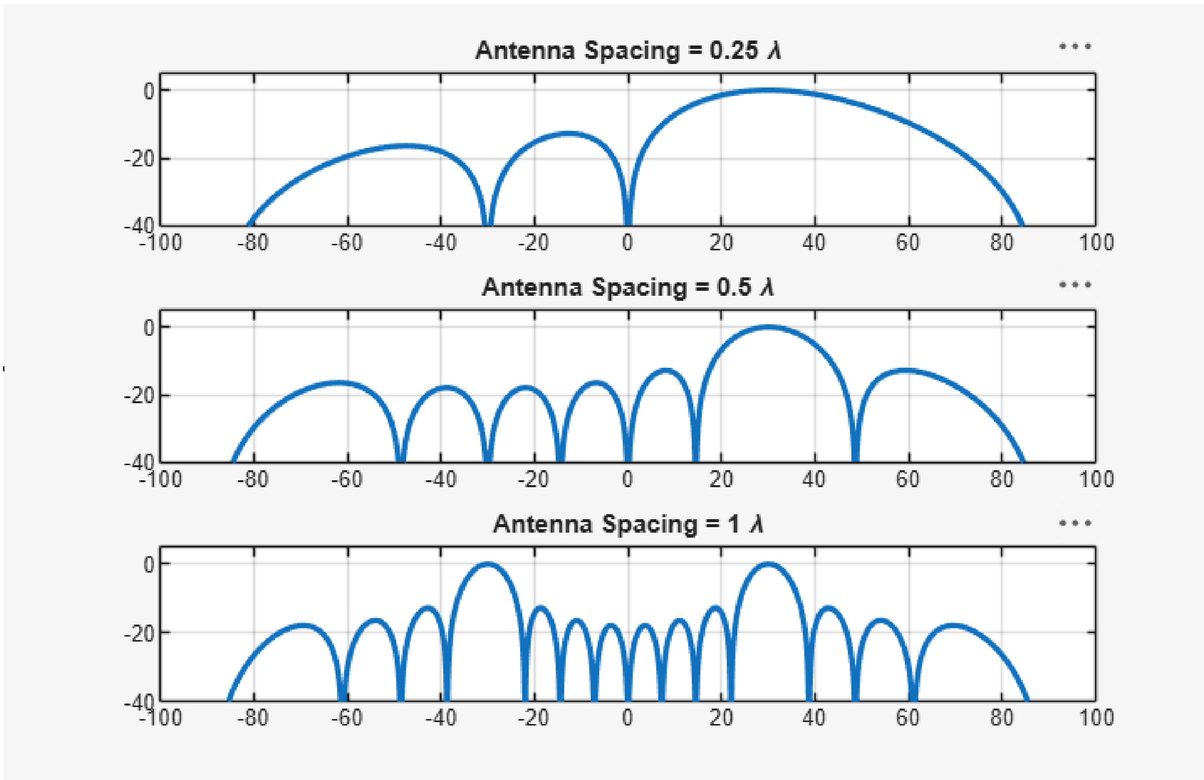
```
noisePower = 1;

SINR = signalPower/(interferencePower+noisePower);

SINR_dB = 10*log10(SINR);

fprintf('SINR = %.2f dB\n',SINR_dB);

%% Effect of Antenna Spacing

spacing = [lambda/4 lambda/2 lambda];

figure;

for i=1:length(spacing)

    d = spacing(i);

    w = exp(-1j*2*pi*d*n'*sind(user_angle)/lambda);

    for k=1:length(theta)

        a = exp(-1j*2*pi*d*n'*sind(theta(k))/lambda);

        AF(k)=abs(w'*a);

    end
```

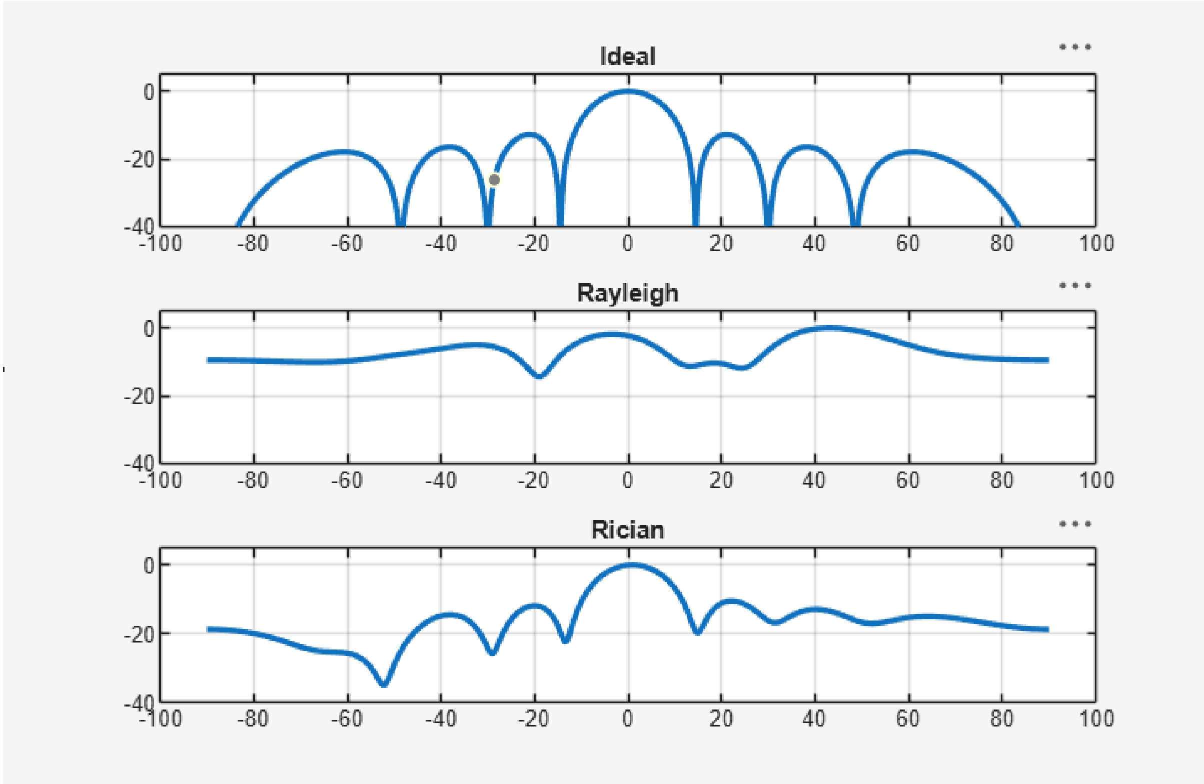
```
    AF=AF/max(AF);

    subplot(3,1,i)
    plot(theta,20*log10(AF),'LineWidth',2)
    grid on
    ylim([-40 5])

    title(['Antenna Spacing = ',num2str(d/lambda),' \lambda'])

end

%% Channel Conditions

channels = {'Ideal','Rayleigh','Rician'};

figure;

for i=1:3

    switch i

        case 1
            h = ones(N,1);

        case 2
            h = (randn(N,1)+1j*randn(N,1))/sqrt(2);

        case 3
```

```
        K=5;
        h = sqrt(K/(K+1))*ones(N,1) + ...
          sqrt(1/(K+1))*((randn(N,1)+1j*randn(N,1))/sqrt(2));
    end

    w = conj(h);

    for k=1:length(theta)

        a = exp(-1j*2*pi*(lambda/2)*n'*sind(theta(k))/lambda);

        AF(k)=abs((w.*a)'*ones(N,1));

    end

    AF=AF/max(AF);

    subplot(3,1,i)
    plot(theta,20*log10(AF),'LineWidth',2)
    grid on
    ylim([-40 5])

    title(channels{i})

end
```

output


Digital Beamforming Pattern
Normalized Gain (dB)
Angle (Degrees)
5
0
-5
-10
-15
-20
-25
-30
-35
-40
-100
-50
0
50
100


Antenna Spacing = 0.25 λ
Antenna Spacing = 0.5 λ
Antenna Spacing = 1 λ
0
-20
-40
-100
-80
-60
-40
-20
0
20
40
60
80
100



Ideal
0
-20
-40
-100
-80
-60
-40
-20
0
20
40
60
80
100
Rayleigh
0
-20
-40
-100
-80
-60
-40
-20
0
20
40
60
80
100
Rician
0
-20
-40
-100
-80
-60
-40
-20
0
20
40
60
80
100